

1) 20

2) 20

3) 20

4) 18

5) 20

98/100

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11/6/19 che 210

HW 10

#1) The standard heat of the reaction: $\text{CaC}_2(\text{s}) + 5 \text{H}_2\text{O}(\text{l}) \rightarrow \text{CaO}(\text{s}) + 2 \text{CO}_2(\text{g}) + 5 \text{H}_2(\text{g})$ is $\Delta H_r^\circ = +69.36 \text{ kJ/mol}$.

(A) Is the reaction exothermic or endothermic at 25°C ? Would you have to heat or cool the reactor to keep the temp constant? What would the temp do if the reactor ran adiabatically? What can you infer about the energy required to break the molecular bonds of the reactants & that released when the product bonds are formed?

• The reaction is Endothermic at 25°C b/c the Heat of Reaction is positive. You would have to heat the reactor to keep the temp constant b/c since it's Endothermic, it is constantly absorbing ^{and lowering T_r} energy. We can infer that the energy absorbed when breaking the bonds is much greater than the energy released when forming the bonds. ✓

(B) Calculate ΔU_r° for this reaction. Briefly explain the physical significance of your calculated value. $R = 8.314 \text{ J/mol}\cdot\text{K}$ $2 \text{ mol}(\text{g}) + 5 \text{ mol}(\text{g})$

$$\Delta U_r^\circ = \Delta H_r^\circ - RT\Delta n_r = (69.36 \text{ kJ/mol}) - (8.314 \text{ J/mol}\cdot\text{K})(298 \text{ K})(7 \text{ mol ggs})$$

$$\Delta U_r^\circ = 52 \text{ kJ/mol} \quad \checkmark \rightarrow \text{Internal energy chemon} \quad \text{1000 J}$$

(C) Suppose you charge 150g of CaC_2 & liquid water into a rigid container at 25°C , heat the container until the calcium carbide reacts completely, and cool the products back down to 25°C , condensing essentially all the unconsumed water. Write & simplify the energy balance equation for this closed constant-volume system & use it to determine the net amount of heat (kJ) that must be transferred to (or from) the reactor. Indicate the direction of heat transfer.

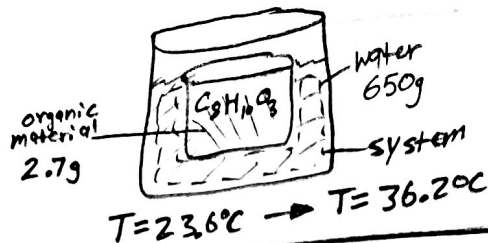
$$\Delta \hat{U} + \Delta \hat{E}_K + \Delta \hat{E}_P = \dot{Q} + \dot{W}_s \rightarrow \Delta \hat{U} = \dot{Q} \rightarrow \dot{Q} = \dot{n} \Delta \hat{U}$$

150g CaC_2 MW = 64.10 g/mol CaC_2

$$\dot{Q} = (52 \text{ kJ/mol}) (150 \text{ g } \text{CaC}_2) \left(\frac{64.10 \text{ g } \text{CaC}_2}{\text{mol}} \right)$$

$$\dot{Q} = 121.875 \text{ kJ} \quad \text{Direction is toward the system}$$

$$\Delta \hat{H}_f^\circ = ?$$



#2) * Consider the change in energy in the inner combustion chamber as being completely due to the chemical reaction & ignore the sensible energy change of the $C_8H_{10}O_3$ & its products. Use the data to calculate an Enthalpy of formation (KJ/g) for the unknown chemical.



$$\Delta \dot{U}_{sys} + \Delta \dot{E}_K + \Delta \dot{E}_P = \dot{Q} + \dot{W}_s \rightarrow \Delta \dot{U}_{sys} = \dot{Q}$$

$$\Delta \dot{U}_{sys} = \dot{Q} = m C_p \Delta T = (650g)(4.18 J/g^\circ C)(36.2^\circ C - 23.6^\circ C)$$

$$\Delta \dot{U}_{sys} = \dot{Q} = 34234.2 J$$

$$* Q = m C_p \Delta T$$

$$* C_p = 4.18 J/g^\circ C \text{ water}$$

$$\Delta U_{sys,in} = -\Delta U_{sys,out} \Rightarrow \Delta U_{sys,in} = -34234.2 J$$

$$\text{We also have } 2.7g \text{ organic material; } \frac{2.7g C_8H_{10}O_3}{154.16g} \times \frac{1 \text{ mol } C_8H_{10}O_3}{1} = 0.0175 \text{ mol } C_8H_{10}O_3$$

$$\Delta U_{sys,in} = \Delta \hat{U}_r^\circ n \Rightarrow \Delta \hat{U}_r^\circ = \frac{\Delta U_{sys,in}}{n} = \frac{(-34234.2 J)}{(0.0175 \text{ mol})} = -1956240 J/mol$$

$$\Delta \hat{U}_r^\circ = \Delta \hat{H}_r^\circ - RT \Delta n_r \Rightarrow \Delta \hat{H}_r^\circ = \Delta \hat{U}_r^\circ + RT(\sum n_{products} - \sum n_{reactants})$$

$$= (-1956240 J/mol) + (8.314 J/mol^\circ K)(298K)(8 \text{ mol} - 9 \text{ mol})$$

$$\Delta \hat{H}_r^\circ = -1958.7 KJ/mol$$

$$\Delta \hat{H}_r^\circ = \sum_i V_i \Delta \hat{H}_{f,i}^\circ = (-1) \Delta \hat{H}_{f,C_8H_{10}O_3}^\circ + (-9) \Delta \hat{H}_{f,O_2}^\circ + (8) \Delta \hat{H}_{f,CO_2}^\circ + (5) \Delta \hat{H}_{f,H_2O}^\circ$$

$$\Delta \hat{H}_{f,C_8H_{10}O_3}^\circ = (8) \Delta \hat{H}_{f,CO_2}^\circ + (5) \Delta \hat{H}_{f,H_2O}^\circ + (-9) \Delta \hat{H}_{f,O_2}^\circ - \Delta \hat{H}_r^\circ$$

* Values from Book

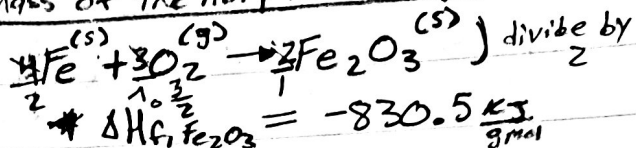
$$\Delta \hat{H}_{f,C_8H_{10}O_3}^\circ = (8)[-393.5 KJ/gmol] + (5)[-285.84 KJ/gmol] - (-1958.7 KJ/mol)$$

$$\Delta \hat{H}_{f,C_8H_{10}O_3}^\circ = -2618.5 KJ/gmol \left(\frac{1 \text{ gmol}}{154.16 g} \right)$$

$$\Delta \hat{H}_{f,C_8H_{10}O_3}^\circ = -16.98 KJ/g$$

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#3) Marketing studies suggest that consumers want a slow heat release of about 100 kJ/h & would like the hotpacks to last about 6 hrs. How much Fe would you put in the hot pack? How much would the mass of the hot pack change over the course of its use (total g & %).



100 kJ/h (6h) = 600 kJ total = Q

$\Delta \hat{H}_{r, \text{Fe}_2\text{O}_3} = (1)(-830.5 \text{ kJ/mol}) = -830.5 \text{ kJ/mol}$

$\Delta \hat{H}_{r, \text{Fe}_2\text{O}_3} = \Delta \hat{U}_r = -830.5 \text{ kJ/mol}$

$Q = n \Delta \hat{U}_r$
 $n = \frac{Q}{\Delta \hat{U}_r} = \frac{(-600 \text{ kJ})}{(-830.5 \text{ kJ/mol})} = 0.722 \text{ mol}$
It's out of system

$0.722 \text{ mol Fe}_2\text{O}_3 \left| \frac{2 \text{ mol Fe}}{1 \text{ mol Fe}_2\text{O}_3} \right| \frac{55.845 \text{ g Fe}}{1 \text{ mol Fe}} =$

80.64 g Fe would be put in the bag

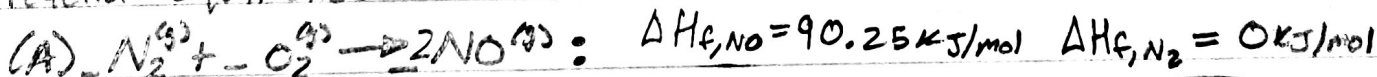
$0.722 \text{ mol Fe}_2\text{O}_3 \left| \frac{159.69 \text{ g Fe}_2\text{O}_3}{1 \text{ mol Fe}_2\text{O}_3} \right| = 115.296 \text{ g Fe}_2\text{O}_3$

Mass change of hot pack: $115.296 \text{ g Fe}_2\text{O}_3 - 80.64 \text{ g Fe} = \mathbf{34.66 \text{ g change}}$

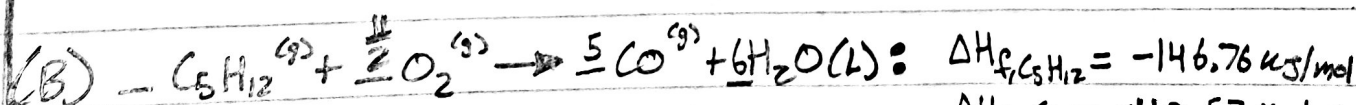
$100 \times \left(\frac{34.66 \text{ g}}{80.64 \text{ g put in}} \right) = \mathbf{43\% \text{ change}}$

#4) Use tabulated heats of formation to determine the standard heats of the following reactions in kJ/mol, letting the stoichiometric coefficient of the first reactant in each reaction equal one.

$\Delta H_{f, \text{O}_2} = 0 \text{ kJ/mol}$



$\Delta H_r = \sum_i \nu_i \Delta H_{f,i} = 2(90.25 \text{ kJ/mol}) - (1)(0) - (1)(0) = \mathbf{180.5 \text{ kJ/mol}}$



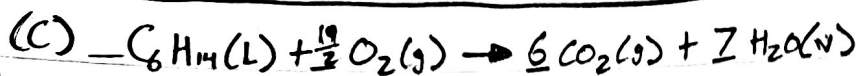
$\Delta H_{f, \text{CO}} = -110.53 \text{ kJ/mol}$

$\Delta H_{f, \text{H}_2\text{O}} = -285.84 \text{ kJ/mol}$

$\Delta H_{f, \text{O}_2} = 0$

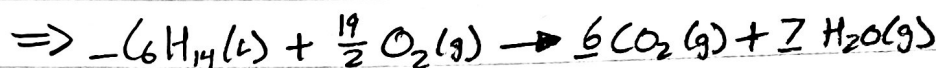
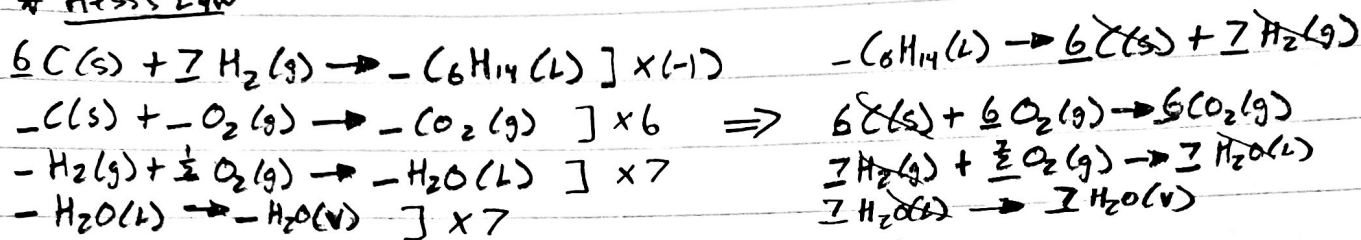
$\Delta H_r = 5(-110.53 \text{ kJ/mol}) + 6(-285.84 \text{ kJ/mol}) - 1(-146.76 \text{ kJ/mol}) - \frac{11}{2}(0) = \mathbf{-2,120.93 \text{ kJ/mol}}$

$$\begin{aligned}\Delta H_f, O_2 &= 0 \\ \Delta H_f, C_6H_6 &= -148.9 \text{ kJ/mol} \\ \Delta H_f, CO_2 &= -393.5 \text{ kJ/mol} \\ \Delta H_f, H_2O &= -241.83 \text{ kJ/mol}\end{aligned}$$

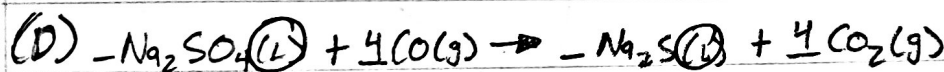


$$\Delta H_r = 6(-393.5 \text{ kJ/mol}) + 7(-241.83 \text{ kJ/mol}) - (-148.9 \text{ kJ/mol}) - \frac{19}{2}(0) = \boxed{-3855.01 \text{ kJ/mol}}$$

* Hess's Law



$$6(-393.5 \text{ kJ/mol}) + 7(-241.83 \text{ kJ/mol}) - (-148.9 \text{ kJ/mol}) - \frac{19}{2}(0) = \boxed{-3885.01 \text{ kJ/mol}}$$



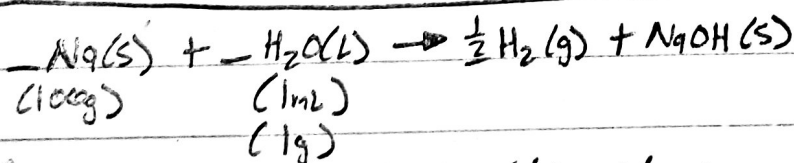
$$\Delta H_r = 1(-323.94 \text{ kJ/mol}) + 4(-393.5 \text{ kJ/mol}) - 1(-1356.38 \text{ kJ/mol}) - 4(-110.53 \text{ kJ/mol}) = \boxed{-99.4 \text{ kJ/mol}}$$

gas

-138.2

$$\begin{aligned}\Delta H_f, CO &= -110.53 \text{ kJ/mol} \\ \Delta H_f, CO_2 &= -393.5 \text{ kJ/mol} \\ \Delta H_f, Na_2SO_4 &= -1356.38 \text{ kJ/mol} \\ \Delta H_f, Na_2S &= -323.94 \text{ kJ/mol}\end{aligned}$$

#5) What else happens? Explain, why it's dangerous to add water to reactive metals, like Sodium



$$\begin{aligned}\Delta H_f, NaOH &= -425.9 \text{ kJ/mol} \\ \Delta H_f, H_2O &= -285.84 \text{ kJ/mol} \\ \Delta H_f, H_2 &= 0 \quad \Delta H_f, Na = 0\end{aligned}$$

$$\Delta H_r = \frac{1}{2}(0) + 1(-425.9 \text{ kJ/mol}) - 1(0) - 1(-285.84 \text{ kJ/mol}) = \boxed{-140 \text{ kJ/mol}}$$

$$\frac{100g H_2O}{18g H_2O} = 0.056 \text{ mol } H_2O \quad \frac{100g Na}{22.9898g Na} = 4.35 \text{ mol } Na$$

$$(4.35 \text{ mol } Na) - (0.056 \text{ mol } H_2O) = 4.29 \text{ mol } Na$$

$$\frac{4.29 \text{ mol } Na}{1 \text{ mol } Na} \times 22.9898g Na = 98.7g Na$$

$$* C_{p, Na} = 28 \text{ J/mol } ^\circ C$$

$$(-140 \text{ kJ/mol})(0.056 \text{ mol}) + (4.29 \text{ mol } Na)(28 \text{ J/mol } ^\circ C) \Delta T \Rightarrow \Delta T = 65^\circ C$$

* This can be dangerous b/c of the high temp change, if the Hydrogen gas can react too violently.